Build a Real AI

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: bigger fires negative positive

1. How long by hand? 240 cards, 300 passes, 1 minute per card.

240 × 300 = _____ cards = _____ minutes ≈ _____ hours ≈ _____ days

2. Six steps to build an AI. Number the steps in order, 1 to 6.

	Train: many passes, fix the weights		Get data
	Split the data: training and test		Test, then use it
	Ask a question		Build the network

Why does the network never train on the test data? _____

3. Read the network.

A blue line is a _____ weight. An orange line is a _____ weight.

A thicker line is a _____ weight. A hidden neuron's square is blue where it _____.

4. Predict, train, and test. Write your prediction **before** each run.

Run	Question	Hidden	My prediction (test %)	Training %	Test %
1	Fighter or Wizard? (2 numbers)	4			
2a	Can it join the club? (2 numbers)	0			
2b	Can it join the club? (2 numbers)	2			
3	Fighter or Wizard? (6 numbers)	16			
4	Where does it live? (6 numbers)	16			

Word bank: data Giants line Minis overfitting

5. What did you see?

In Run 2a, the network had no hidden layer. It could only draw one _____.

In Run 2b, one hidden neuron fires only for _____. The other fires for everything except _____.

In Run 3, the training score went up, but the test score went down. The network learned the training examples by heart. This is called _____.

In Run 4, the network wasn't better than guessing. Why? _____

An AI can only learn what is in its _____.

6. Talk about it. What if someone used an AI like this to decide who gets a job?

A problem could be _____

_____

_____

Go Further (Day 2). Ask the network about 3 creatures from the cards.

Creature	The network said	Really	Right?	Test creature?

Fighter or Wizard? (all 6 numbers). Test score with 0 hidden: _____ 4: _____ 16: _____

Is the biggest network always the best? _____ Why? _____